Comparing DeltaQuant to Related Works

Ahmad Fraij

Using Prefix Cache For FreeAct Comparison

No-cache results

Model	Setting	Method	HumanEval	GSM8K	MATH500	Avg.	Δ%
	BF16	Base (ours, Fast-dLLM)	41.46	78.09	38.0	52.52	–
	BF16	Base (FreeAct paper)	43.90	76.80	38.40	53.03	–

Prefix cache results

Model	Setting	Method	HumanEval	GSM8K	MATH500	Avg.	Δ%
	BF16	Base (ours, Fast-dLLM)	43.29	77.10	38.4	52.93	–
	BF16	Base (FreeAct paper)	43.90	76.80	38.40	53.03	–

Issues with Integrating DeltaQuant with Prefix Cache

- No-cache: every timestep involves a forward pass on entire sequence (all activation distributions have same shape (seq_length,channel dim))
- Prefix-cache: each diffusion block involves two differently shaped forward passes:
  - A warm up pass on the whole sequence to recompute the cached values
  - Multiple passes that do not include recomputation of cached parts of the sequence
- Calculating the delta between two activation matrices with different shapes will not work

Combining DeltaQuant with Prefix Cache

- At the first timestep store the full activation inputs and outputs in the cache
- In subsequent timesteps, only the matching portion of the cache is updated
- Within that matching portion, take the difference between the new activation and what's cached for that slice, quantize just that difference, run it through the linear layer, and add it onto the cached output for that slice.
- At scheduled anchor points, the matching portions of input and output cache are directly updated with the FP16 values from that timestep.
- This allows for delta quantization between activations of different shapes.

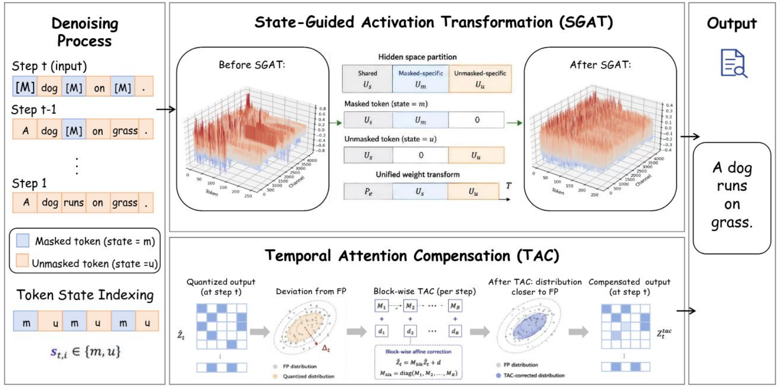
Comparison with FreeAct using PrefixCache

LLaDA-8B-Instruct accuracy: our reproduction via Fast-dLLM (prefix cache, real FlashAttention) vs. FreeAct's own reported numbers, at W4A4.

Model	Setting	Method	HumanEval	GSM8K	MATH500	Avg.	Δ%
LLaDA-8B-Instruct	BF16	Base	43.29	77.10	38.4	52.93	–
	W4A4	QuaRot	32.32	68.23	28.4	42.98	↓18.80%
		DeltaQuaRot	34.15	72.02	31.0	45.72	↓13.62%
		DeltaQuaRot (best: clip 0.999, 16 anchors)	39.02	72.18	32.80	48.00	↓9.31%
		DuQuant	36.59	73.84	36.4	48.94	↓7.54%
		DeltaDuQuant (best: 1 anchor)	42.07	76.04	36.0	51.37	↓2.95%
		FreeAct	42.68	75.74	34.60	51.01	↓3.63%

StarQuant

- Activation ranges varying significantly between masked tokens and unmasked tokens in a sequence
- StarQuant has two main methods:
  - State guided activation transformation (SGAT): splitting the tokens according to their states, then transform them into different subspaces. The outputs are then scattered back to original token order
  - Temporal attention compensation (TAC): deals with temporal error accumulation by compensating the quantized attention representation before the output projection layer



Comparison with DLLMQuant and StarQuant (Nocache)

LLaDA-8B accuracy: our reproduction via Fast-dLLM vs. DLLMQuant's and STaR-Quant's own reported numbers, at W4A4.

Model	Setting	Method	HumanEval	GSM8K	Avg.	Δ%
LLaDA-8B	BF16	Base	32.93	66.64	49.78	–
	W4A4	QuaRot	26.22	61.71	43.97	↓11.69%
		DeltaQuaRot	29.27	61.56	45.41	↓8.78%
		DeltaQuaRot (best: clip 0.999, 4 anchors)	33.54	61.41	47.47	↓4.64%
		DuQuant	25.00	57.16	41.08	↓17.49%
		DeltaDuQuant	29.88	57.85	43.87	↓11.89%
		DeltaDuQuant (best: clip 0.995, 16 anchors)	31.10	58.00	44.55	↓10.52%
		DLLMQuant^{++}	28.92	56.25	42.59	↓14.46%
		STaR-Quant	35.98	57.29	46.63	↓6.33%

Dream Results

Dream-v0-Instruct-7B BF16 baseline: our reproduction via Fast-dLLM (prefix cache, nochat) vs. FreeAct's own reported numbers.

Model	Setting	Method	HumanEval	GSM8K	MATH500	Avg.
Dream-Instruct-7B	BF16	FreeAct (reported)	56.10	68.16	42.20	55.49
		Ours (reproduction)	55.49	73.62	41.4	56.84

Dream-v0-Instruct-7B accuracy: our reproduction via Fast-dLLM (prefix cache, nochat), at W4A4.

Model	Setting	Method	HumanEval	GSM8K	MATH500	Avg.	Δ%
Dream-Instruct-7B	BF16	Base	55.49	73.62	41.4	56.84	–
	W4A4	QuaRot (noclip)	34.15	56.03	29.8	39.99	↓29.63%
		DeltaQuaRot (1 anchor)	42.07	21.61	39.80	34.49	↓39.31%
		DeltaQuaRot (best: clip 0.95, 4 anchors)	46.95	36.16	41.60	41.57	↓26.86%
		DuQuant	3.66	30.02	11.0	14.89	↓73.80%
		DeltaDuQuant (1 anchor)	32.32	48.52	26.80	35.88	↓36.87%
		DeltaDuQuant (best: clip 0.95, 16 anchors)	32.93	55.65	27.60	38.73	↓31.87%
		FreeAct	50.00	68.84	32.60	50.48	↓11.19%

Outline for Implementation Section

- Activation Analysis for LLaDA Model across Timesteps
  - (Relative) Mean Absolute Difference
  - (Relative) Max Absolute Difference
  - Kurtosis
  - Signal-to-Noise Ratio (SNR)
  - Cosine Similarity
- DeltaQuant: Initial Mathematical Formulation
  - Error Accumulation in the Initial Formulation
  - Self-Correcting DeltaQuant Formulation
- Combining DeltaQuant with Rotation-Based Quantization Techniques
  - DeltaQuaRot
  - DeltaDuQuant
- Reanchoring Techniques for DeltaQuant
  - Uniform Reanchoring
  - Offline Calibrated Reanchoring
  - Online Reanchoring
- Applying DeltaQuant to the Fast-dLLM Prefix Cache Method
  - Combining Reanchoring with Prefix Caching

Outline for Results Section

- Experimental Setup
  - Math/Coding Benchmarks
  - LLaDA/Dream Models
  - Hyperparameters Used
- Results for Initial DeltaQuant Method
- Results for Self-Correcting DeltaQuant Method
  - LLaDA-Base
  - LLaDA-Instruct
  - Dream-Base
  - Dream-Instruct
- Results for DeltaQuant with Fast-dLLM Prefix Cache
- Comparison to Related Work
  - FreeAct
  - DLLMQuant
  - StarQuant
- Limitations
  - Extra Memory Requirements for Cache Storage

Plan for Next Week

- Work on implementation and results section for the report